Spain data center decree: what the hourly-correlation surcharges actually cost

Ex post scoring of our solved portfolios under Art. 10.4 and Art. 11 of the draft Real Decreto of 26 Oct 2026. Internal note, 8 Nov 2026. Model and portfolios: `pypsalabs/spain-datacenter-decree`, 2025 fleet, Spain-wide PPA siting.

What Art. 10 allows (the 80% every-hour rule is a surcharge, not a hard constraint)

- Art. 9.1: in every hour at least 80% of consumption backed by same-hour renewable generation from contracted plants, incl. associated storage.
- Art. 10.4: hourly shortfall is 0 at ≥80% backing, 100% at zero backing, linear in between. Monthly shortfall S = arithmetic mean of the hourly shortfalls.
- Surcharge on the month's grid tariffs (peajes + cargos, all unit prices): $S < 5\%$: 10%; 5–10%: 15%; 10–20%: 30%; 20–50%: 70%; ≥50%: 100%. Plus 10 pp per additional consecutive year of non-compliance.
- Art. 10.6: surcharge revenue counts as income for system charges, i.e. lowers cargos for other consumers. A transfer, not a resource cost.
- Art. 11.1.b: loss of access permit only if the "porcentaje de horas de incumplimiento" is ≥20% for five consecutive years; not defined whether this is the share of failing hours or the Art. 10 interpolated mean.

How we score it ex post (no new optimisation)

- Take the solved hourly dispatch of each portfolio: PPA generation, DC battery charge/discharge, DC consumption (flat 500 MW; 4 GW as scaling check).
- Hourly backing = PPA generation + battery discharge − battery charging, capped at consumption. Art. 9.4 storage rule (grey share of grid charging deducted from later production) implemented literally (though it changes nothing, the batteries barely charge from the grid).
- Shortfall per hour as in Art. 10.4, monthly mean, tier per month, surcharge = tier × that month's tariff bill.
- Procurement cost (€/MWh, PPAs + battery + grid purchases − sales) from the published `results/scenarios.csv`. All-in = procurement + surcharge.
- Dispatch is not re-optimised to dodge tiers, so the surcharge is an upper bound for each portfolio.

Portfolio (500 MW)	Failing hours	Mean shortfall	Months per tier 10/15/30/70/100 %	Surcharge €/MWh	Surcharge M€/yr	Procurement €/MWh	All-in €/MWh
Hourly 80% on annual average	35%	15%	0/2/8/2/0	1.3	5.8	54.1	55.4
Hourly 90% on annual average	18%	7%	3/5/4/0/0	0.8	3.3	62.3	63.1
Strict 80% every hour (decree)	0%	0%	none	0.0	0.0	93.3	93.3
Transition P1 60% (DT1, to 2028)	47%	25%	0/0/3/9/0	2.5	10.9	57.1	59.6
Transition P1 70% / P2 60% (DT1)	32%	13%	0/4/7/1/0	1.2	5.3	63.1	64.3
No policy (no PPAs, full surcharge)	100%	100%	0/0/0/0/12	4.0	17.5	72.8	76.8

Surcharge base. Tariff bill of a flat 500 MW load on 6.4TD (six-period access tariff, connection ≥145 kV) at 2026 unit prices: peajes from CNMC Resolución 18 Dec 2025 (BOE-A-2025-26348) plus cargos set by MITECO. Power term 15.8 €/kW·yr summed over P1–P6 (contracted power in all periods) = 1.8 €/MWh; energy term 11.28/6.06/2.60/1.29/0.40/0.23 €/MWh in P1–P6, mainland calendar of Circular 3/2020 = 2.2 €/MWh. Total 4.0 €/MWh, so the 100% tier doubles the grid tariff and costs 4.0 €/MWh. A 6.3TD connection (72.5–145 kV) roughly doubles the base. Art. 11 trigger (≥20%): binary reading hits hourly 80%, both transitions and no policy; interpolated reading hits transition P1 60% and no policy only; hourly 90% escapes both, two points from the binary line. At 4 GW surcharges are the same per MWh within 0.1; all-in gap to strict 36 (80%) and 28 (90%) €/MWh.

Takeaways

- The surcharge cannot force the overbuild: max 4 €/MWh (about 6 by year five) against a 38 €/MWh gap between strict compliance and the 80%-average portfolio. Paying the full surcharge with no PPAs at all is still 16 €/MWh cheaper than strict compliance.
- The only teeth is Art. 11 permit loss, and its metric is ambiguous. De facto rule for a rational operator: stay under 20% failing hours and pay about 1 €/MWh.